



**United International University**  
**Department of Computer Science and Engineering**  
**CSE 3812: Artificial Intelligence Laboratory**  
**Assignment 1**

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## Problem Description

In the storm-ravaged city of **Pacifica**, emergency response teams are deploying drones to deliver critical supplies across a chaotic urban grid. The city is represented as an  **$N \times N$  grid of sectors**, where each sector has a **travel cost** reflecting environmental challenges such as strong winds, debris, or flooded zones. Some sectors are completely impassable due to collapsed buildings or ongoing rescue operations and are marked as **obstacles**.

Your task is to program a drone to deliver a **portable defibrillator**, a vital medical device, from the **Relief Hub** at sector  **$(S_x, S_y)$**  to a stranded medical team at sector  **$(D_x, D_y)$** . The drone can move in **eight directions** (north, south, east, west, and the four diagonals) and must avoid obstacles while finding the path with the lowest total travel cost.

To achieve this, you will implement the ***A search algorithm***<sup>\*</sup>, which uses both the **actual cost from the start** and a **heuristic estimate** of the destination. The heuristic will be the **Euclidean distance**, calculating the straight-line distance to the medical team's location, ensuring the drone navigates efficiently through the city.

Your goal is to compute the path with the lowest total cost, ensuring the drone delivers the defibrillator quickly and safely to save lives.

## Requirements

Implement the  $A^*$  search algorithm to:

1. Calculate the **total cost** of the optimal path from  $(S_x, S_y)$  to  $(D_x, D_y)$ .
2. Provide the **sequence of sectors** (path) that the drone should follow.

## Input Representation:

- **City Grid:** A 2D matrix **Grid[N][N]** where:
  - ➡ Each element is a **non-negative integer** representing the travel cost of that sector.
  - ➡ Sectors with **-1** indicate obstacles (impassable).
- **Start Position:**  $(S_x, S_y) \rightarrow$  the drone's starting sector.
- **Destination Position:**  $(D_x, D_y) \rightarrow$  the medical team's sector.

## Drone Movement

- The drone can move in **eight directions**:
  - ➡ Cardinal: up (north), down (south), left (west), right (east).
  - ➡ Diagonal: up-left, up-right, down-left, down-right.
- **Cost Calculation**: The cost of moving to a neighboring sector is the **travel cost of the destination sector**. For example, moving from (x, y) to (x+1, y+1) incurs the cost of  $\text{Grid}[x+1][y+1]$ .

## Heuristic Function

- Use the **Euclidean distance** to estimate the cost from the current sector (x, y) to the destination (Dx, Dy):

$$h(x,y) = \sqrt{(x - Dx)^2 + (y - Dy)^2}$$

- This heuristic is admissible (it never overestimates the true cost) and helps guide the drone efficiently toward the goal.

## Output Explanation

The output consists of two components:

1. **Total Cost**:
  - ➡ This is the sum of the travel costs of the sectors along the optimal path, starting from the first move.
  - ➡ **Important Note**: The cost of the starting sector (Sx, Sy) is **not included** in the total cost because the drone begins there without incurring a travel cost. Only the costs of the sectors it moves into are summed.
  - ➡ **Example**: If the path is  $(0, 0) \rightarrow (0, 1) \rightarrow (1, 2)$  with costs [1, 1, 1], the total cost is  $1 + 1 = 2$  (excluding the cost at (0, 0)).
2. **Optimal Path**:
  - ➡ A sequence of coordinates (as tuples) representing the sectors the drone should traverse, from (Sx, Sy) to (Dx, Dy), inclusive.
  - ➡ **The path must avoid obstacles (sectors with -1) and reflect the lowest possible total cost.**
  - ➡ **Example**:  $(0, 0) \rightarrow (0, 1) \rightarrow (1, 2) \rightarrow (2, 3) \rightarrow (3, 3)$  shows the full route from start to destination.

If no valid path exists (due to obstacles blocking all routes), the output should indicate this, typically with a total cost of **None** and **an empty path []**.

**Example:**

| Sample Input 1                                                                                                                                                                                                                                                                 | Sample Output 1 |   |    |    |   |    |   |   |   |   |   |   |   |    |   |   |                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|---|----|----|---|----|---|---|---|---|---|---|---|----|---|---|----------------------------------------------------------------------------------------------|
| <div>Grid:</div> <table><tr><td>1</td><td>1</td><td>1</td><td>-1</td></tr><tr><td>1</td><td>-1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>-1</td><td>1</td><td>1</td></tr></table> <div>Sx,Sy=(0,0)<br/>Dx,Dy=(3,3)</div> | 1               | 1 | 1  | -1 | 1 | -1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | -1 | 1 | 1 | <div>Optimal Cost: 4</div> <div>Optimal Path:</div> <div>(0,0)→(0,1)→(1,2)→(2,3)→(3,3)</div> |
| 1                                                                                                                                                                                                                                                                              | 1               | 1 | -1 |    |   |    |   |   |   |   |   |   |   |    |   |   |                                                                                              |
| 1                                                                                                                                                                                                                                                                              | -1              | 1 | 1  |    |   |    |   |   |   |   |   |   |   |    |   |   |                                                                                              |
| 1                                                                                                                                                                                                                                                                              | 1               | 1 | 1  |    |   |    |   |   |   |   |   |   |   |    |   |   |                                                                                              |
| 1                                                                                                                                                                                                                                                                              | -1              | 1 | 1  |    |   |    |   |   |   |   |   |   |   |    |   |   |                                                                                              |

**Explanation:**

1. The drone starts at (0, 0) (cost not included).
2. Moves to [0, 1] (cost 1), [1, 2] (cost 1), [2, 3] (cost 1), and [3, 3] (cost 1).
3. Total cost = 1 + 1 + 1 + 1 = 4.
4. Obstacles at (0, 3), (1, 1), and (2, 1) are avoided.

| Sample Input 2                                                                                                                                                                                                                                                                 | Sample Output 2 |   |    |    |   |    |   |   |   |   |   |   |   |    |   |   |                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|---|----|----|---|----|---|---|---|---|---|---|---|----|---|---|----------------------------------------------------------------------------------------------|
| <div>Grid:</div> <table><tr><td>2</td><td>3</td><td>1</td><td>-1</td></tr><tr><td>1</td><td>-1</td><td>4</td><td>2</td></tr><tr><td>1</td><td>2</td><td>3</td><td>1</td></tr><tr><td>3</td><td>-1</td><td>2</td><td>1</td></tr></table> <div>Sx,Sy=(0,0)<br/>Dx,Dy=(3,3)</div> | 2               | 3 | 1  | -1 | 1 | -1 | 4 | 2 | 1 | 2 | 3 | 1 | 3 | -1 | 2 | 1 | <div>Optimal Cost: 6</div> <div>Optimal Path:</div> <div>(0,0)→(1,0)→(2,1)→(3,2)→(3,3)</div> |
| 2                                                                                                                                                                                                                                                                              | 3               | 1 | -1 |    |   |    |   |   |   |   |   |   |   |    |   |   |                                                                                              |
| 1                                                                                                                                                                                                                                                                              | -1              | 4 | 2  |    |   |    |   |   |   |   |   |   |   |    |   |   |                                                                                              |
| 1                                                                                                                                                                                                                                                                              | 2               | 3 | 1  |    |   |    |   |   |   |   |   |   |   |    |   |   |                                                                                              |
| 3                                                                                                                                                                                                                                                                              | -1              | 2 | 1  |    |   |    |   |   |   |   |   |   |   |    |   |   |                                                                                              |

## Basic Functions Needed

To implement the A\* algorithm, you'll need several helper functions. You can find the template here: [A\\_star template](#)

## Submission Guidelines

1. **File Format:** Submit your solution as a .ipynb (Jupyter Notebook) or .py (Python script) file.
2. **File Naming:** Name your file using your Student ID (011201274\_A\*.ipynb or 011201274\_A\*.py).
3. **Resources:** You may use online resources, but ensure you fully understand your code for the viva. Failure to explain your implementation during the viva will reduce your marks.
4. **Bonus Opportunities:**
  - ➡ **File Input/Output (+2 marks):** Earn bonus marks if you implement reading the grid, start, and goal positions from an input file ( .txt) and writing the output (cost and path) to an output file instead of hardcoding them. Include a sample input file with your submission.
  - ➡ **Matplotlib Visualization (+2 marks):** Gain additional marks by using the Matplotlib library to display the grid visually and the optimal path ( marking obstacles, start, goal, and path). Include the visualisation code in your submission.

## Marks Distribution

- Input + Output: 5 marks
- Heuristic Implementation: 7 marks
- Viva: 8 marks
- Subtotal: 20 marks
- Bonus (File I/O & Matplotlib): Up to 4 marks
- Total Possible: 24 marks